Lecture Scribe CSE400

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Go through this whole material and Generate a lecture scribe suitable for exam preparation. For every concept, definition, and assumption, explicitly state the reasoning and logical dependencies as presented in the context. Use only the provided context. Do not introduce outside intuition, new examples, or external explanations. The final output must allow a student to reliably revise and reconstruct the core arguments and solutions of the lecture without needing additional resources.

Below is a **faithful, exam-ready lecture scribe** reconstructed **strictly from the provided lecture material (Lecture 8, CSE400)**.
All definitions, formulas, identities, examples, and logical steps are presented **exactly as they appear in the slides and handwritten annotations**, with **explicit reasoning and dependencies**, and **no external intuition or additions**.

This scribe is designed so that a student can **revise and reconstruct the lecture arguments and problem-solving flow independently**.

Source: Lecture slides and annotations, *CSE400 – Fundamentals of Probability in Computing, Lecture 8*

CSE400 – Lecture 8

Gaussian, Uniform, Exponential, Laplace, and Gamma Random Variables

(Exam-Ready Lecture Scribe)

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1. Preliminaries: Moments and Central Moments

1.1 Definition: Central Moment

For a continuous random variable X with PDF $f_X(x)$, the nth **central moment** is defined as:

$$\mathrm{E}[(X-\mu_X)^n] = \int \quad (x-\mu_X)^n f_X(x)\,dx$$

This definition depends on:

- The existence of the PDF $f_X(x)$
- The mean $\mu_X = \mathrm{E}[X]$

1.2 Special Cases of Central Moments

Case $n = 0$

$$\mathrm{E}[(X-\mu_X)^0] = \mathrm{E}[1] = 1$$

Case $n = 1$

$$\mathrm{E}[X-\mu_X] = \mathrm{E}[X] - \mu_X = 0$$

This follows directly from the definition of mean.

Case $n = 2$**: Variance**

$$\sigma_X^2 = \mathrm{E}[(X-\mu_X)^2]$$

Expanding:

$$\mathrm{E}[X^2 - 2\mu_X X + \mu_X^2] = \mathrm{E}[X^2] - 2\mu_X \mathrm{E}[X] + \mu_X^2$$

Using $\mathrm{E}[X] = \mu_X$:

$$\sigma_X^2 = \mathrm{E}[X^2] - \mu_X^2$$

1.3 Higher-Order Central Moments

Skewness ($n = 3$)

$$C_s = \frac{\mathrm{E}[(X-\mu_X)^3]}{\sigma_X^3}$$

Interpretation (as stated in lecture):

- $C_s > 0$: PDF is **right-skewed**

- $C_s < 0$: PDF is **left-skewed**

Kurtosis ($n = 4$)

$$C_k = \mathrm{E}[(X - \mu_X)^4]$$

A **large value** indicates:

- The random variable has a **large peak near the mean**

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2. Linearity Properties of Expectation

Theorem 1: Linearity with Constants

For constants a and b:

$$\mathrm{E}[aX + b] = a\mathrm{E}[X] + b$$

This follows from the linearity of integration.

Theorem 2: Linearity over Sums

For any function decomposed as:

$$g(x) = g_1(x) + g_2(x) + \cdots + g_N(x)$$

Then:

$$\mathrm{E}[\sum_{k=1}^{N} g_k(X)] = \sum_{k=1}^{N} \mathrm{E}[g_k(X)]$$

Expectation via PDF

$$\mathrm{E}[X] = \int \quad x f_X(x)\, dx$$

This expression is the foundational definition used throughout the lecture.

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3. Gaussian Random Variable

3.1 Definition

A random variable X is **Gaussian** if its PDF can be written as:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp(-\frac{(x-m)^2}{2\sigma^2})$$

Notation:

$$X \sim \mathrm{N}(m, \sigma^2)$$

Where:

- $m = \mathrm{E}[X]$ (mean)
- $\sigma^2 = \mathrm{Var}(X)$

3.2 Standard Normal Distribution

When:

$$m = 0, \quad \sigma^2 = 1$$

The distribution is called the **standard normal distribution**.

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4. Standard Functions Associated with Gaussian RVs

4.1 Error Function

$$\mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2}\, dt$$

4.2 Complementary Error Function

$$\mathrm{erfc}(x) = 1 - \mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \int_x^\infty e^{-t^2}\, dt$$

4.3 Φ-Function (CDF of Standard Normal)

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-t^2/2}\, dt$$

4.4 Q-Function (Gaussian Tail Function)

$$Q(x) = \frac{1}{\sqrt{2\pi}} \int_{x}^{\infty} e^{-t^2/2}\, dt$$

Key Identity:

$$Q(x) = 1 - \Phi(x)$$

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5. Relation Between Φ-Function and Q-Function

5.1 Evaluating the CDF of a Gaussian RV

Starting from:

$$F_X(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi\sigma^2}} \exp(-\frac{(y-m)^2}{2\sigma^2})\, dy$$

Apply the substitution:

$$t = \frac{y-m}{\sigma}$$

Then:

$$F_X(x) = \Phi(\frac{x-m}{\sigma})$$

5.2 Evaluating Tail Probabilities

$$\Pr(X > x) = \int_{x}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp(-\frac{(y-m)^2}{2\sigma^2})\, dy$$

After substitution:

$$\Pr(X > x) = Q(\frac{x-m}{\sigma})$$

5.3 Key Identities (Explicitly Listed)

$$Q(x) = 1 - \Phi(x)$$

$$F_X(x) = 1 - Q(\frac{x - m}{\sigma})$$

Lecture Note:

- To evaluate the **CDF**, evaluate the **Φ-function**
- The **Q-function** is more natural for probabilities of the form $\Pr(X > x)$

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6. Worked Example (Gaussian)

Given

$$f_X(x) = \frac{1}{\sqrt{8\pi}} \exp(-\frac{(x+3)^2}{8})$$

Step 1: Identify Parameters

Comparing with:

$$\frac{1}{\sqrt{2\pi\sigma^2}} \exp(-\frac{(x-m)^2}{2\sigma^2})$$

We identify:

$$m = -3, \quad \sigma^2 = 4 \Rightarrow \sigma = 2$$

Step 2: Use Gaussian Relations

$$F_X(x) = \Phi(\frac{x - m}{\sigma})$$

$$\Pr(X > x) = Q(\frac{x - m}{\sigma})$$

Step 3: Required Probabilities

1. $\Pr(X \leqslant 0)$
   Left-tail probability → use Φ-function
2. $\Pr(X > 4)$
   Right-tail probability → use Q-function
3. $\Pr(|X + 3| < 2)$
   Convert to interval:

$$-2 < X + 3 < 2 \Rightarrow -5 < X < -1$$

   Evaluate using:

$$F_X(-1) - F_X(-5)$$

4. $\Pr(|X - 2| > 1)$
   Convert to union of intervals and evaluate using CDF and Q-function as appropriate

Lecture Hint:

- Use **Q(x)** for right-tail probabilities
- Use **Φ(x)** for left-tail probabilities

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7. Summary of Logical Flow

1. Define moments → derive variance, skewness, kurtosis
2. Establish linearity of expectation
3. Define Gaussian PDF and parameters
4. Introduce standard functions (erf, erfc, Φ, Q)
5. Derive CDF and tail probabilities using substitution
6. Apply identities systematically in problem solving

End of Lecture Scribe

(Strictly reconstructed from provided lecture material, with no external additions.)

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the context. Use only the provided context. Do not introduce outside intuition, new examples, or external explanations. The final output must allow a student to reliably revise and reconstruct the core arguments and solutions of the lecture without needing additional resources.

Below is a **faithful, exam-ready lecture scribe** reconstructed **only from the provided lecture material (Lecture 8, CSE400)**.
All concepts, definitions, formulas, assumptions, and solution steps are written **exactly in the logical order and with the dependencies used in the lecture**, including what is assumed, what is derived, and why each step follows.
No outside intuition, no new examples, and no external explanations are introduced.

Source: *CSE400 – Fundamentals of Probability in Computing, Lecture 8*

CSE400 – Lecture 8

Gaussian, Uniform, Exponential, Laplace, and Gamma Random Variables

Exam-Ready Lecture Scribe

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1. Moments and Central Moments

1.1 Definition: Central Moment

Let X be a continuous random variable with PDF $f_X(x)$ and mean $\mu_X = \mathrm{E}[X]$.

The nth **central moment** of X is defined as:

$$\mathrm{E}[(X - \mu_X)^n] = \int (x - \mu_X)^n f_X(x)\, dx$$

Logical dependencies:

- Requires the PDF $f_X(x)$
- Requires the mean μ_X

- Expectation is defined using integration over the PDF

1.2 Evaluation of Low-Order Central Moments

Case $n = 0$

$$E[(X - \mu_X)^0] = E[1] = 1$$

This follows because any nonzero quantity raised to power 0 equals 1.

Case $n = 1$

$$E[X - \mu_X] = E[X] - \mu_X = \mu_X - \mu_X = 0$$

Reasoning:
Uses linearity of expectation and the definition of the mean.

Case $n = 2$: Variance

$$\sigma_X^2 = E[(X - \mu_X)^2]$$

Expanding the square:

$$(X - \mu_X)^2 = X^2 - 2\mu_X X + \mu_X^2$$

Taking expectation:

$$E[X^2] - 2\mu_X E[X] + \mu_X^2$$

Using $E[X] = \mu_X$:

$$\sigma_X^2 = E[X^2] - \mu_X^2$$

1.3 Higher-Order Central Moments

Skewness ($n = 3$)

$$C_s = \frac{E[(X - \mu_X)^3]}{\sigma_X^3}$$

Interpretation given in lecture:

- $C_s > 0$: PDF is right-skewed
- $C_s < 0$: PDF is left-skewed

Kurtosis ($n = 4$)

$$C_k = \mathrm{E}[(X - \mu_X)^4]$$

Lecture statement:
A large value of kurtosis indicates that the random variable has a **large peak near the mean**.

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2. Properties of Expectation

2.1 Linearity with Constants

For constants a and b:

$$\mathrm{E}[aX + b] = a\mathrm{E}[X] + b$$

Dependency:
Follows from linearity of integration.

2.2 Linearity over Sums

If a function is written as:

$$g(x) = g_1(x) + g_2(x) + \cdots + g_N(x)$$

Then:

$$\mathrm{E}[\sum_{k=1}^{N} g_k(X)] = \sum_{k=1}^{N} \mathrm{E}[g_k(X)]$$

2.3 Expectation Using PDF

$$\mathrm{E}[X] = \int x f_X(x)\, dx$$

This is the base definition used throughout the lecture.

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3. Gaussian Random Variable

3.1 Definition

A random variable X is **Gaussian** if its PDF can be written as:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp(-\frac{(x-m)^2}{2\sigma^2})$$

Where:

- $m = \mathrm{E}[X]$ (mean)
- $\sigma^2 = \mathrm{Var}(X)$

Notation:

$$X \sim \mathrm{N}(m, \sigma^2)$$

3.2 Standard Normal Distribution

If:

$$m = 0, \quad \sigma^2 = 1$$

Then X is called a **standard normal random variable**.

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4. Standard Forms Associated with Gaussian RVs

4.1 Error Function

$$\mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2}\, dt$$

4.2 Complementary Error Function

$$\mathrm{erfc}(x) = 1 - \mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \int_x^\infty e^{-t^2}\, dt$$

4.3 Φ-Function (CDF of Standard Normal)

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-t^2/2}\, dt$$

4.4 Q-Function (Gaussian Tail Function)

$$Q(x) = \frac{1}{\sqrt{2\pi}} \int_{x}^{\infty} e^{-t^2/2}\, dt$$

Identity explicitly stated:

$$Q(x) = 1 - \Phi(x)$$

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5. Relation Between Φ-Function and Q-Function

5.1 Evaluating the CDF of a Gaussian RV

Start from the definition:

$$F_X(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi\sigma^2}} \exp(-\frac{(y-m)^2}{2\sigma^2})\, dy$$

Substitute:

$$t = \frac{y-m}{\sigma}$$

Then:

$$F_X(x) = \Phi(\frac{x-m}{\sigma})$$

5.2 Evaluating Tail Probabilities

$$\Pr(X > x) = \int_{x}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp(-\frac{(y-m)^2}{2\sigma^2})\, dy$$

After substitution:

$$\Pr(X > x) = Q(\frac{x-m}{\sigma})$$

5.3 Key Identities (As Listed)

$$Q(x) = 1 - \Phi(x)$$

$$F_X(x) = 1 - Q(\frac{x - m}{\sigma})$$

Lecture notes:

- To evaluate the **CDF**, use the **Φ-function**
- The **Q-function** is more natural for probabilities of the form $\Pr(X > x)$

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6. Worked Gaussian Example

Given PDF

$$f_X(x) = \frac{1}{\sqrt{8\pi}} \exp(-\frac{(x+3)^2}{8})$$

Step 1: Identify Parameters

Compare with:

$$\frac{1}{\sqrt{2\pi\sigma^2}} \exp(-\frac{(x-m)^2}{2\sigma^2})$$

Thus:

$$m = -3, \quad \sigma^2 = 4, \quad \sigma = 2$$

Step 2: Use Gaussian Relations

$$F_X(x) = \Phi(\frac{x - m}{\sigma})$$

$$\Pr(X > x) = Q(\frac{x - m}{\sigma})$$

Step 3: Required Probabilities

1. $\Pr(X \leqslant 0)$
   Left-tail probability → use Φ-function
2. $\Pr(X > 4)$
   Right-tail probability → use Q-function
3. $\Pr(|X + 3| < 2)$
   Convert to:

$$-5 < X < -1$$

   Evaluate as:

$$F_X(-1) - F_X(-5)$$

4. $\Pr(|X - 2| > 1)$
   Convert to union of intervals and evaluate using Φ and Q as appropriate

Lecture hint:
Use **Q(x)** for right-tail probabilities and **Φ(x)** for left-tail probabilities.

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7. Logical Flow of the Lecture

1. Define central moments and evaluate special cases
2. Derive variance from second central moment
3. Introduce skewness and kurtosis
4. Establish linearity properties of expectation
5. Define Gaussian random variable and notation
6. Introduce erf, erfc, Φ, and Q functions
7. Derive CDF and tail probabilities using substitution
8. Apply identities to solve Gaussian probability problems

End of Lecture Scribe
(Strictly based on the provided lecture material, suitable for direct exam revision.)

Go through this whole material and Generate a lecture scribe suitable for exam preparation. For every concept, definition, and assumption, explicitly state the reasoning and logical dependencies as presented in

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Every concept, definition, identity, and solution step is written **in the same logical order, with the same assumptions and dependencies used in the lecture**.
Nothing is added, generalized, or explained using outside intuition.
Topics that appear only in the outline but **do not appear in the actual provided content** are **not expanded**.

Source: *CSE400 – Fundamentals of Probability in Computing, Lecture 8*

CSE400 – Lecture 8

Gaussian Random Variable and Associated Functions

(Exam-Ready Lecture Scribe)

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1. Moments and Central Moments

1.1 Definition: Central Moment

Let X be a continuous random variable with PDF $f_X(x)$ and mean $\mu_X = \mathrm{E}[X]$.

The nth **central moment** is defined as:

$$\mathrm{E}[(X - \mu_X)^n] = \int (x - \mu_X)^n f_X(x)\, dx$$

Dependencies explicitly used:

- Expectation is defined via integration over the PDF.
- The mean μ_X must exist.

- The expression depends only on $f_X(x)$ and μ_X.

1.2 Evaluation of Central Moments for Specific Orders

Case $n = 0$

$$E[(X - \mu_X)^0] = E[1] = 1$$

This follows directly from the definition of expectation.

Case $n = 1$

$$E[X - \mu_X] = E[X] - \mu_X = \mu_X - \mu_X = 0$$

Reasoning used:
Linearity of expectation and the definition of the mean.

Case $n = 2$: Variance

$$\sigma_X^2 = E[(X - \mu_X)^2]$$

Expand the square:

$$(X - \mu_X)^2 = X^2 - 2\mu_X X + \mu_X^2$$

Taking expectation:

$$E[X^2] - 2\mu_X E[X] + \mu_X^2$$

Substituting $E[X] = \mu_X$:

$$\sigma_X^2 = E[X^2] - \mu_X^2$$

1.3 Higher-Order Central Moments

Skewness ($n = 3$)

$$C_s = \frac{E[(X - \mu_X)^3]}{\sigma_X^3}$$

Lecture interpretation:

- $C_s > 0$: PDF is right-skewed
- $C_s < 0$: PDF is left-skewed

Kurtosis ($n = 4$)

$$C_k = \mathrm{E}[(X - \mu_X)^4]$$

Lecture statement:

A large value implies the random variable has a **large peak near the mean**.

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2. Properties of Expectation

2.1 Linearity with Constants

For constants a and b:

$$\mathrm{E}[aX + b] = a\mathrm{E}[X] + b$$

Dependency:

Follows directly from linearity of integration.

2.2 Linearity over Sums

If a function is written as:

$$g(x) = g_1(x) + g_2(x) + \cdots + g_N(x)$$

Then:

$$\mathrm{E}[\sum_{k=1}^{N} g_k(X)] = \sum_{k=1}^{N} \mathrm{E}[g_k(X)]$$

2.3 Expectation via PDF

$$\mathrm{E}[X] = \int x f_X(x)\, dx$$

This is the foundational definition used in all derivations.

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3. Gaussian Random Variable

3.1 Definition

A random variable X is **Gaussian** if its PDF is:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp(-\frac{(x-m)^2}{2\sigma^2})$$

Where:

- $m = \mathrm{E}[X]$
- $\sigma^2 = \mathrm{Var}(X)$

Notation:

$$X \sim \mathrm{N}(m, \sigma^2)$$

3.2 Standard Normal Distribution

If:

$$m = 0, \quad \sigma^2 = 1$$

Then the distribution is the **standard normal distribution**.

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4. Standard Functions Associated with Gaussian RVs

4.1 Error Function

$$\mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2}\, dt$$

4.2 Complementary Error Function

$$\mathrm{erfc}(x) = 1 - \mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \int_x^\infty e^{-t^2}\, dt$$

4.3 Φ-Function (CDF of Standard Normal)

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-t^2/2}\, dt$$

4.4 Q-Function (Gaussian Tail Function)

$$Q(x) = \frac{1}{\sqrt{2\pi}} \int_{x}^{\infty} e^{-t^2/2}\, dt$$

Identity stated in lecture:

$$Q(x) = 1 - \Phi(x)$$

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5. Relation Between Φ-Function and Q-Function

5.1 Evaluating the CDF of a Gaussian RV

Start from the CDF definition:

$$F_X(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi\sigma^2}} \exp(-\frac{(y-m)^2}{2\sigma^2})\, dy$$

Substitute:

$$t = \frac{y-m}{\sigma}$$

Then:

$$F_X(x) = \Phi(\frac{x-m}{\sigma})$$

5.2 Evaluating Tail Probabilities

$$\Pr(X > x) = \int_{x}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp(-\frac{(y-m)^2}{2\sigma^2})\, dy$$

After substitution:

$$\Pr(X > x) = Q(\frac{x-m}{\sigma})$$

5.3 Key Identities (Explicitly Listed)

$$Q(x) = 1 - \Phi(x)$$

$$F_X(x) = 1 - Q(\frac{x - m}{\sigma})$$

Lecture notes:

- Φ-function is used to evaluate the **CDF**
- Q-function is more natural for probabilities of the form $\Pr(X > x)$

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6. Gaussian Example (Fully from Lecture)

Given PDF

$$f_X(x) = \frac{1}{\sqrt{8\pi}} \exp(-\frac{(x+3)^2}{8})$$

Step 1: Identify Parameters

Compare with the Gaussian PDF:

$$\frac{1}{\sqrt{2\pi\sigma^2}} \exp(-\frac{(x-m)^2}{2\sigma^2})$$

Thus:

$$m = -3, \quad \sigma^2 = 4, \quad \sigma = 2$$

Step 2: Gaussian Relations Used

$$F_X(x) = \Phi(\frac{x - m}{\sigma})$$

$$\Pr(X > x) = Q(\frac{x - m}{\sigma})$$

Step 3: Probabilities to Be Evaluated

1. $\Pr(X \leqslant 0)$
   Left-tail probability → Φ-function
2. $\Pr(X > 4)$
   Right-tail probability → Q-function
3. $\Pr(|X + 3| < 2)$
   Convert:

$$-2 < X + 3 < 2 \Rightarrow -5 < X < -1$$

   Evaluate as:

$$F_X(-1) - F_X(-5)$$

4. $\Pr(|X - 2| > 1)$
   Converted into tail probabilities and evaluated using Φ and Q as appropriate

Lecture hint explicitly stated:
Use **Q(x)** for right-tail probabilities and **Φ(x)** for left-tail probabilities.

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7. Logical Structure of the Lecture

1. Define central moments
2. Derive variance from second central moment
3. Introduce skewness and kurtosis
4. Establish linearity of expectation
5. Define Gaussian random variable
6. Introduce erf, erfc, Φ, and Q
7. Derive CDF and tail probabilities via substitution
8. Apply identities to solve Gaussian probability problems

End of Lecture Scribe
(Fully constrained to the provided material and suitable for direct exam revision without external resources.)